Pranshu Gupta

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Professional Experience

- **Microsoft SuperIntelligence** / *Member of Technical Staff / Redmond, WA* — *Jul 2026 – Present*
  - Productionized Multi-LoRA serving for **MAI-Code-1-Flash** by delivering an inference engine image for Azure Foundry.
  - Benchmarked Multi-LoRA scaling, profiled GPU bottlenecks with PyTorch/Nsight, and tuned CSGMV LoRA kernel.
  - De-risked **MAI-Thinking-1** LoRA serving by uncovering incompatibilities with data parallelism and speculative decoding. Partnered with OSS maintainers to land upstream compatibility fixes for both serving modes.
- **Microsoft AI** / *Member of Technical Staff / Redmond, WA* — *Aug 2025 – Jul 2026*
  - Served as **technical owner** for inference on Copilot, shaping architectural decisions across model integration, and safety.
  - Engineered the inference pipeline for **GPT-5.1 instant and reasoning models** on Copilot at launch, with meta-prompting for personality alignment, delivering a **30% improvement in personality evals** for the reasoning path.
  - Scoped, designed, and built **GPU-based safety classifier** endpoints for the Chat Completions API: non-streaming endpoints that return classification logits for harm scoring on prompts and responses.
  - Designed and shipped an **incremental chain of thought summarization** system: buffers streaming reasoning chunks and progressively surfaces summaries in the UI, keeping it responsive while protecting raw model reasoning as IP.
  - Reduced inference costs by eliminating non-essential classifiers (**reducing GPU footprint by 20%**) and resolving a token overhead inefficiency in Chat Completions API (**reducing input tokens by 25%** and **TTFT by 30ms**).
  - Defined and instrumented LLM API **traffic classification** dimensions (product, client app, inference intent, license tier) across MAI and M365 Copilot; built Grafana dashboards informing capacity planning and cost attribution across both orgs.
- **Microsoft Azure AI** / *Senior Software Engineer / Redmond, WA* — *Jan 2025 – Aug 2025*
  - Designed and built a distributed **token-based rate limiter** using Tiktoken for precise token counting, replacing a length-based estimator that had 219% average error; implemented pub-sub quota state synchronization across instances.
  - Designed and built the inference pipeline for an intelligent **model router**: injecting per-request model selection based on cost/complexity signals before each LLM call, reducing per-inference cost by **55%**.
- **Microsoft Azure Core** / *Senior Software Engineer / Redmond, WA* — *Mar 2024 – Jan 2025*
  - Served as **technical area owner** for resource lifecycle management in Azure Resource Manager.
  - Mentored a new-grad engineer through weekly 1:1s; they now independently own a critical component in control plane.
  - Designed and shipped a multi-bucket **token-based background job throttling** system with retry-storm protection, preventing noisy-neighbor throttling in Azure Control Plane background-job infrastructure.
  - Single-handedly designed and shipped **async operation URI signing**; ran office hours with resource provider teams to guide API surface changes, adopted by **242 service teams** with **220+ enforcing it**.
- **Microsoft Azure Core** / *Software Engineer II / Redmond, WA* — *Jun 2020 – Mar 2024*
  - Designed and shipped **async operation callbacks** to replace polling, cutting control plane polling traffic by **94%**.
  - Led **Azure region teardown** design and implementation end-to-end: halted new resource creation, resolved subscription/resource group hierarchies, and migrated live resources to healthy regions with zero data loss.
  - Designed and built a **health-driven deployment pipeline** for ARM control-plane manifests: staged multi-region rollouts with post-step RP health checks that halt or rollback on degradation.
- **Microsoft Core Services** / *Software Engineer / Hyderabad, India* — *Jun 2017 – Aug 2019*
  - Modernized a data enrichment platform to Azure Function Apps (**$4K/month** savings); integrated Cognitive Services for data standardization, and built a bulk-processing pipeline with **40–50x throughput** improvement.

Technical Skills

- **AI/ML & Inference**: LLM inference/serving, prompt engineering, model evaluations, safety orchestration, RAG
- **Languages & Systems**: Python, .NET, C++, KQL, JavaScript, HTML, CSS, Kubernetes, Redis, Grafana
- **Cloud & Infrastructure**: Azure (Cosmos DB, ARM Templates, Service Bus, DevOps), AWS (Bedrock, S3, ECR)

Publications & Writing

- *Runtime Optimizations for LLMs* (2026). Production notes on inference optimizations for latency/cost-efficient LLM serving.
- *The Observer Effect on Social Media* — ACM CHI (2024). Causal inference and time series study on longitudinal Facebook data from 300+ participants to quantify how monitoring awareness alters online behavior.
- *Mental Health Impact of Active-Shooter Drills* — Nature (2021). Applied ML classifiers and interrupted time series analysis to 54M social media posts across 114 schools to measure psychological and civic impact.

Education

- **Georgia Institute of Technology** / *M.S. Computer Science (Machine Learning) / Atlanta, GA* — *May 2020*
- **Indian Institute of Technology, Kanpur** / *B.Tech, C.S. and Eng. / Kanpur, India* — *May 2017*